

Path Metrics Computations & Theories

Based on `calculate_path_metrics_gpx.py`

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1 Distance Calculation (Spherical & Euclidean Theory)

2D Ground Distance (d_{2d}): Uses the Haversine Formula, which is the standard trigonometric theory for calculating the great-circle distance between two points on a sphere (the Earth).

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cos(\phi_2) \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (1)$$

$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a}) \quad (2)$$

$$d_{2d} = R \cdot c \quad (3)$$

Where $R = 6371000$ meters (Earth's radius), ϕ is latitude, and λ is longitude.

3D Travel Distance (d_{3d}): Uses the Pythagorean Theorem to account for elevation changes, ensuring the distance includes the physical hypotenuse of traveling up/down a hill.

$$d_{3d} = \sqrt{d_{2d}^2 + \Delta z^2} \quad (4)$$

Where $\Delta z = z_2 - z_1$ (elevation difference).

2 Topographical Computations

Total Ascent/Descent: Iterates through every segment between GPX points. If the elevation difference (Δz) is positive, it adds it to the total

ascent. If negative, the absolute value is added to total descent.

$$\text{Ascent} = \sum_{\Delta z_i > 0} \Delta z_i \quad (5)$$

$$\text{Descent} = \sum_{\Delta z_i < 0} |\Delta z_i| \quad (6)$$

Max Slope (S_{max}): Calculated by finding the slope of every individual segment and tracking the highest absolute value found.

$$\text{Slope}_i = \left(\frac{\Delta z_i}{d_{2d,i}} \right) \times 100\% \quad (7)$$

$$S_{max} = \max_i (|\text{Slope}_i|) \quad (8)$$

Overall Gradient: The average slope of the entire path from start to finish.

$$\text{Gradient} = \left(\frac{\sum \Delta z}{\sum d_{2d}} \right) \times 100\% \quad (9)$$

3 Kinematic Penalty Theory (Speed & Travel Time)

Instead of a complex non-linear curve (like Tobler's function), this script uses a **Linear Slope Penalty Theory**.

Slope Penalty (P): Linear rule where base speed reduces by 2% for every 1% of absolute overall gradient. Capped at a maximum penalty of 80% (0.8).

$$P = \min(|\text{Gradient}| \times 0.02, 0.8) \quad (10)$$

Adjusted Speed (V_{adj}):

$$V_{adj} = V_{base} \times (1 - P) \quad (11)$$

Where $V_{base} = 8.33 \text{ m/s}$ (approx. 30 km/h, typically for vehicle travel).

Travel Time with Slope (T_{slope}):

$$T_{slope} = \frac{\sum d_{3d}}{V_{adj}} \quad (12)$$

To get minutes: $T_{min} = \frac{T_{slope}}{60}$